

Regional Integration as a Stumbling Block: Gravity Evidence from EU–ACP Trade

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1 Research Question and Motivation

A central unresolved question in the political economy of trade is whether preferential regional trade agreements act as *building blocks* toward broader multilateral liberalization or as *stumbling blocks* that divert trade away from non-member partners (Bhagwati, 1993). This paper applies that question to the relationship between the African, Caribbean, and Pacific (ACP) group and the European Union—one of the most institutionally complex North–South trade relationships in the world.

Since the signing of the Cotonou Agreement in 2000, ACP countries have been negotiating a shift from non-reciprocal Lomé preferences toward reciprocal Economic Partnership Agreements (EPAs) with the EU. Simultaneously, ACP sub-regions have deepened their own intra-bloc integration through seven Regional Economic Communities (RECs): ECOWAS, SADC, EAC, COMESA, Central Africa, CARIFORUM, and the Pacific Islands Forum (PIF). Whether this regional integration supports or undermines ACP–EU trade is an open empirical question with direct implications for EPA design and the ongoing post-Cotonou negotiations.

2 Data and Methodology

Data

Bilateral trade flows are drawn from the BACI HS92 dataset (CEPII, V202601) for 1995–2021. Gravity covariates—geographic distance, common language, and colonial ties—come from the CEPII Gravity Dataset (V202211). GDP and population data are sourced from the World Bank WDI, supplemented with manual entries for Cook Islands and Niue. The panel comprises 78 ACP countries paired with each of the 27 EU member states over 27 years, yielding 56,862 potential dyad-years; the estimation sample contains 45,647 observations after conditioning on data availability.

The key explanatory variable is the **intra-REC trade share** (IT share), defined at the country–year level as intra-bloc exports plus imports as a share of total country trade. For SACU member states, missing or zero intra-bloc flows are imputed using the non-SACU SADC annual mean. A

secondary measure, **IT intensity**, normalizes the IT share by the bloc’s share of world trade to account for differences in bloc size.

Empirical Strategy

I estimate PPML gravity models following Santos Silva and Tenreyro (2006), which accommodates heteroskedasticity and the large share of zero trade flows in the panel. The baseline specification is:

$$X_{ijt} = \exp(\alpha_{it} + \alpha_j + \alpha_t + \beta_1 \text{IT}_{jt} + \beta_2 \text{EPA}_{jt} + \gamma \mathbf{Z}_{ij}) \cdot \varepsilon_{ijt}$$

where X_{ijt} is bilateral trade between EU exporter i and ACP importer j in year t ; α_{it} are exporter–year fixed effects; α_j are ACP-country fixed effects; α_t are year fixed effects; IT_{jt} is the intra-REC trade share; EPA_{jt} is an indicator for an EPA being in force; and $\mathbf{Z}_{ij}$ includes log distance, common language, and colonial tie.

Importer–year fixed effects—the standard structural gravity solution—would absorb both IT share and EPA indicators, which vary only at the ACP-country–year level. The chosen fixed-effects structure instead preserves identification while controlling for common time-period shocks. Identification comes from *within-country* variation in the IT share over time, exploited across the 27 EU partner observations per country–year. I additionally estimate an interaction specification ($\text{EPA} \times \text{IT share}$) to test whether EPA coverage moderates the stumbling-block effect. Standard errors are clustered at the dyad level throughout.

3 Preliminary Results

Main Findings

Table 1 reports the core estimates. The baseline PPML model (Column 1) yields the expected gravity coefficients: log distance -1.729^{***} and common language $+1.107^{***}$. Column 2 introduces the IT share, producing a coefficient of -0.564^* (s.e. 0.322), consistent with the stumbling-block hypothesis. At the sample mean IT share of 11.6%, this implies a 6.3% reduction in bilateral EU–ACP trade; at the 90th percentile (25.5%), the estimated reduction reaches 13.4%.

Column 3 adds the $\text{EPA} \times \text{IT share}$ interaction. The interaction coefficient is $+1.048^{**}$ (s.e. 0.477), indicating that EPAs significantly attenuate the stumbling-block effect for covered country-years. A full sign reversal would require an IT share exceeding 0.60, which lies outside the sample support. Column 4 replaces the IT share with IT intensity, yielding -0.006^{***} (s.e. 0.002), confirming robustness to the choice of integration measure. Column 5 presents OLS estimates on log trade (zero pairs excluded), which produces a larger coefficient of -1.478^{***} , directionally consistent with the PPML baseline.

Table 1: Intra-REC Trade Integration and EU–ACP Bilateral Trade

	Baseline PPML (1)	IT Share PPML (2)	IT $\times$ EPA PPML (3)	IT Intensity PPML (4)	IT Share OLS (5)
ln(Distance)	−1.729*** (0.609)	−1.786*** (0.644)	−1.786*** (0.644)	−1.767*** (0.642)	−1.173*** (0.239)
Common Language	1.107*** (0.218)	1.118*** (0.223)	1.118*** (0.223)	1.115*** (0.222)	0.674*** (0.125)
EPA (in force)	−0.018 (0.049)	−0.019 (0.049)	−0.139* (0.075)	−0.044 (0.045)	−0.174*** (0.046)
Intra-REC Trade Share		−0.564* (0.322)	−0.625* (0.320)		−1.478*** (0.236)
EPA $\times$ IT Share			1.048** (0.477)		
IT Intensity				−0.006*** (0.002)	
Exporter $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
ACP Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	46,160	45,647	45,647	45,317	42,305

Notes: Pair-level clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Panel: 78 ACP countries $\times$ EU-27 $\times$ 1995–2021. Columns (1)–(4) estimated via PPML; column (5) via OLS on $\log(\text{trade})$ excluding zero-trade pairs.

Mechanisms and Heterogeneity

Decomposing trade by direction reveals that the stumbling-block effect is driven entirely by *EU exports to ACP markets* (coefficient -1.027^{***}), while ACP exports to the EU are unaffected (-0.059 , insignificant). This asymmetry is consistent with intra-REC import substitution displacing European goods rather than a reorientation of ACP export supply chains.

REC-level subsamples yield uniformly negative IT share coefficients, with five of six interpretable estimates statistically significant. Point estimates range from -0.75 (Central Africa, insignificant) to -3.02^{***} (EAC). The Caribbean (CARIFORUM) estimate of -2.327^{**} is among the largest, reflecting both deep intra-bloc integration and strong EPA coverage. Within SADC, staggered EPA entry across member states generates within-REC identification, with an EPA coefficient of $+0.408^{***}$.

Robustness

The stumbling-block finding is stable across a range of checks: negative binomial PML (-0.458), sample trimmed to 1998 onward (-1.336^{***}), exclusion of SACU members (-1.590^{***}), and exclusion of dominant traders (South Africa, Nigeria). Excluding the SACU imputation *strengthens* rather than weakens the main estimate, ruling out the imputation as a driver of the result. The OLS $\log(1 + y)$ sign reversal is a known artefact of the near-zero transformation and is not interpreted substantively.

4 Conclusion

This paper provides systematic gravity evidence that intra-regional trade integration among ACP countries acts as a stumbling block to EU–ACP trade. The effect is concentrated in the direction of EU exports, consistent with an import substitution mechanism. Economic Partnership Agreements attenuate, but do not eliminate, this trade diversion. These findings suggest that the benefits of deepening ACP regional integration depend critically on its coordination with preferential trade arrangements—sequential implementation is unlikely to replicate the trade gains achievable under simultaneous advancement of both regional and North–South liberalization.